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 SURVEY

Beyond the Leaderboard: A Survey of the Science of Evaluation, Benchmarking, and Methodologies for Large Language Models

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ABSTRACT The field of artificial intelligence is grappling with an evaluation crisis. Despite the proliferation of benchmarks and the relentless pursuit of state-of-the-art scores on public leaderboards, confidence in what these metrics truly measure is eroding. Large Language Models (LLMs) now consistently outperform humans on established benchmarks. However, their real-world utility is often undermined by brittleness, a lack of common sense, and a tendency to generate harmful or factually incorrect content. This paper argues that this disconnect stems from treating evaluation as a competitive afterthought rather than a rigorous metrological science. We contend that the current paradigm, characterized by static benchmarks and a narrow focus on accuracy, is fundamentally ill-equipped to assess the complex, multifaceted capabilities of modern LLMs. This survey aims to provide a comprehensive reference for the science of LLM evaluation by making four principal contributions. First, we offer a historical analysis of evaluation, tracing its evolution from traditional lexical-overlap metrics to the massive, holistic frameworks of today. Second, we introduce a systematic taxonomy of evaluation methodologies, classifying them into five key paradigms: static benchmarks, dynamic and adaptive evaluation, interactive and agentic evaluation, human-in-the-loop evaluation, and model-based evaluation (LLM-as-a-Judge). Third, we conduct a critical analysis of the pathologies plaguing the field, with a deep focus on the existential threat of data contamination, the perverse incentives of the leaderboard obsession, and the pervasive mismatch between benchmark performance and true capability. Finally, we propose a new conceptual framework, the Benchmark Lifecycle, to professionalize the design, validation, deployment, monitoring, and retirement of evaluation instruments. By framing evaluation as a scientific discipline, we aim to move beyond the leaderboard and build a more robust, reliable, and trustworthy foundation for measuring progress in artificial intelligence.

INDEX TERMS Large language models, evaluation, benchmarking, natural language processing, artificial intelligence, metrology, data contamination, AI safety, emergent capabilities.

I. INTRODUCTION: THE CRISIS OF CONFIDENCE IN LLM EVALUATION

A. THE EVALUATION PARADOX: MORE BENCHMARKS, LESS CERTAINTY

The field of artificial intelligence (AI) is experiencing an unprecedented paradox. On the one hand, the capabilities of Large Language Models (LLMs) are advancing at a

brehtaking pace, with models now achieving superhuman scores on a wide range of standardized tests. Headlines announce that models can “pass” complex professional examinations in medicine, law, and business, suggesting a rapid approach towards human-level competence [22]. On the other hand, a significant crisis of confidence is emerging within the research community and among practitioners [28]. The very benchmarks designed to measure progress are increasingly seen as unreliable, gameable, and disconnected from the qualities we desire in AI systems: robustness,

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fairness, factual accuracy, and real-world utility. We have more ways to measure LLMs than ever before, yet we understand less and less what those measurements mean. This disconnect is the central “evaluation crisis” of our time [21]. It displays the growing gap between leaderboard performance and practical capability. A model may achieve a near-perfect score on a multiple-choice question-answering dataset but fail at simple, real-world reasoning tasks that a child could solve [42]. This evaluation-capability mismatch is not only an academic concern; it has profound implications for the safe and responsible deployment of AI. As LLMs are integrated into high-stakes domains like healthcare and finance, a flawed understanding of their true abilities, fostered by misleading benchmark scores, poses a significant risk [5]. The community is grappling with the realization that our traditional evaluation metrics and methods are becoming ineffective, if not entirely outdated, in the face of powerful generative models [23].

B. A HISTORICAL TRAJECTORY: FROM GLUE TO THE PROLIFERATION ERA

The history of evaluation in Natural Language Processing (NLP) can be viewed as a three-act narrative, as illustrated in Figure 1. Each act is defined by the dominant evaluation paradigm and the models that challenged it.

Act I: Consolidation and Task-Specificity. The pre-LLM era was characterized by task-specific metrics and a move toward consolidated benchmarks. Metrics like BLEU (Bilingual Evaluation Understudy) for machine translation [38] and ROUGE (Recall-Oriented Understudy for Gisting Evaluation) for summarization [31] provided quantitative, automated ways to measure performance on narrow tasks. The key development of this era was the General Language Understanding Evaluation (GLUE) benchmark [46]. GLUE aggregated nine diverse NLU tasks, creating a single-number metric that galvanized the community and incentivized the development of models with transferable knowledge, most notably BERT [15].

Act II: Saturation and the Transformer Revolution. The introduction of large-scale Transformer models marked the second act. Models like GPT-2 and GPT-3 demonstrated a shocking degree of generalization, performing a wide array of tasks with zero or few examples provided in the prompt [8], [40]. This new paradigm of in-context learning rapidly “saturated” existing benchmarks. GLUE, once a formidable challenge, was surpassed by models within a year of its release [45]. The community responded reactively by creating SuperGLUE, a “stickier” benchmark with more difficult and diverse tasks designed to provide more headroom for these powerful new models [45]. This period established a cat-and-mouse dynamic: as models scaled, benchmarks that once seemed comprehensive were quickly rendered obsolete.

Act III: Proliferation and Fragmentation. We are currently in the third act, an era of massive proliferation and methodological fragmentation. The limitations of task-specific leaderboards led to the creation of massive,

multitask benchmarks designed to test broad world knowledge and reasoning. MMLU (Measuring Massive Multitask Language Understanding) set a new standard for difficulty by testing expert-level knowledge across 57 subjects [22]. BIG-Bench followed, a massive collaborative effort to create tasks believed to be far beyond the capabilities of existing models [44]. Simultaneously, a parallel movement toward holistic evaluation emerged, as demonstrated by Stanford’s HELM (Holistic Evaluation of Language Models), which assesses models not only on accuracy but also on a multi-metric suite that includes fairness, robustness, and efficiency [30]. This era is also defined by an explosion of domain-specific benchmarks for code (HumanEval) [10], medicine (MedQA), and agentic behavior (SWE-bench) [26]. While this proliferation has provided more specialized tools, it has also created a paradox of choice, where inconsistent protocols and a lack of standardization make it challenging to compare results meaningfully [28].

C. THE CENTRAL THESIS: EVALUATION AS A METROLOGICAL SCIENCE

This survey argues for a fundamental paradigm shift in how we approach the assessment of LLMs. The reactive cycle of building benchmarks only to see them broken must end. Instead, we must elevate evaluation from a competitive sport of leaderboard-chasing to a strict **metrological science**, the science of measurement itself. This perspective requires that we view our benchmarks not as static finish lines, but as scientific instruments, each with its own specifications, error bars, and operational limits.

A metrological approach to evaluation is built on several core principles. First, it requires a principled, systematic design process in which the capabilities being measured are explicitly defined and justified. Second, it demands rigorous validation, including establishing strong human baselines, conducting statistical power analyses, and stress-testing for robustness. Third, and most critically, it necessitates a lifecycle perspective, recognizing that every benchmark has a finite period of utility and requires active management from its creation to its eventual, planned retirement. This paper aims to lay the groundwork for this new science of evaluation.

D. CONTRIBUTIONS AND STRUCTURE OF THIS SURVEY

This survey aims to be the definitive reference for the science of LLM evaluation. To achieve this, we make four primary contributions:

- 1) **A Historical Analysis of Evaluation:** We trace the evolution of LLM evaluation through a three-act narrative, from traditional lexical-overlap metrics to today’s massive, holistic frameworks, contextualizing the current crisis of confidence (Section IV).
- 2) **A Comprehensive Taxonomy:** we introduce a comprehensive, systematic classification of modern LLM evaluation methodologies, organizing the fragmented landscape into a coherent framework that enables

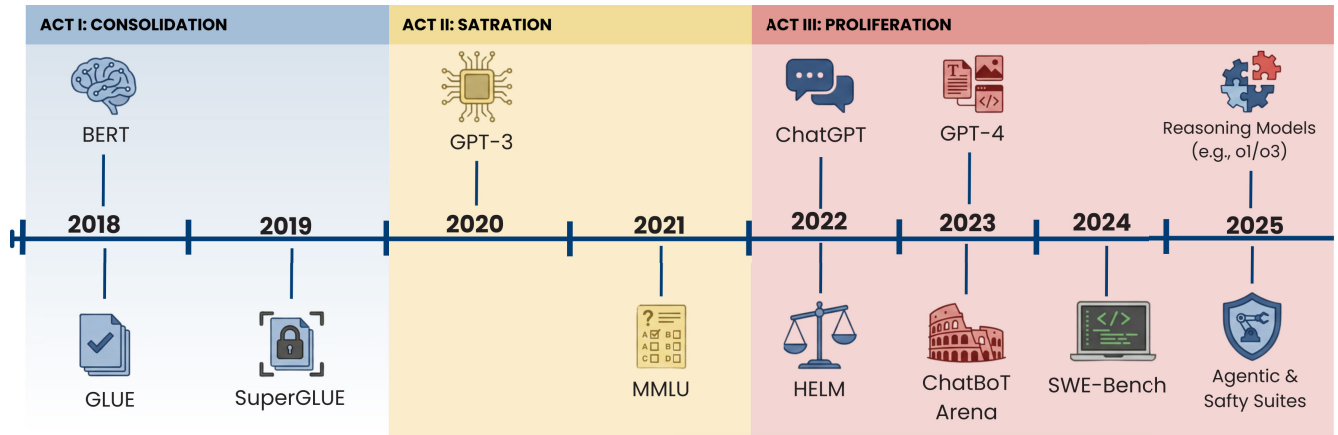


FIGURE 1. The three acts of evaluation history. The timeline illustrates the “cat-and-mouse” dynamic in which new model capabilities (top) rapidly saturate existing benchmarks (bottom), prompting the creation of new evaluation paradigms.

principled comparison and selection of evaluation strategies (Section V).

- 3) **The Benchmark Lifecycle Framework:** We propose a novel conceptual model for the end-to-end management of benchmarks. This framework provides a structured process for design, validation, deployment, contamination monitoring, and retirement, treating benchmarks as professional-grade scientific instruments (Section VI).
- 4) **A Critical Analysis of Pathologies:** We conduct a deep and critical examination of the systemic failures that undermine the field’s credibility. This includes a thorough analysis of the leaderboard obsession and Goodhart’s Law, the growing chasm between benchmark scores and real-world utility, and an extensive investigation into the data contamination crisis (Section VI).
- 5) **A Holistic, Multi-Dimensional Vision:** Throughout this paper, we champion a move away from monolithic accuracy scores. We advocate for a multi-faceted evaluation paradigm that embraces metrics for safety, fairness, efficiency, and robustness as first-class citizens in the assessment of model quality (Section VIII).
- 6) **A Forward-Looking Research Agenda:** We conclude by identifying the most pressing open research questions and providing concrete recommendations for researchers, practitioners, policymakers, and benchmark creators to foster a more mature and reliable science of evaluation (Section X).

The structure of this paper follows these contributions, beginning with a historical review, presenting our core frameworks, critically analyzing current pathologies, surveying advanced paradigms, and finally, looking toward the future. By synthesizing the vast literature and providing new organizing principles, this work endeavors to resolve the current crisis of confidence and set a new, more rigorous course for the field. To complement this three-act narrative, Table 1 summarizes key milestones in LLM evaluation

and the paradigm shifts they enabled, highlighting how new models repeatedly forced the community to revise its measurement tools.

II. RESEARCH OBJECTIVES

This survey addresses four critical gaps in the current literature on LLM evaluation:

A. PRIMARY RESEARCH OBJECTIVES

- 1) **RO1: Systematic Classification** - To develop a comprehensive taxonomy of LLM evaluation methodologies that enables researchers to navigate the fragmented landscape of evaluation approaches and make principled selections based on their specific requirements.
- 2) **RO2: Lifecycle Framework Development** - To propose and validate a novel Benchmark Lifecycle Framework that treats evaluation instruments as scientific tools requiring systematic management from design through retirement, addressing the current ad-hoc approach to benchmark creation and maintenance.
- 3) **RO3: Pathology Analysis** - To conduct a critical analysis of systemic failures in current evaluation practices, particularly examining (a) the contamination crisis affecting benchmark validity, (b) the evaluation-capability mismatch documented across domains, and (c) the perverse incentives created by leaderboard-focused research culture.
- 4) **RO4: Metrological Foundation** - To establish evaluation as a metrological science by introducing principles from measurement science to LLM assessment, moving beyond competitive benchmarking toward rigorous, reproducible measurement practices.

B. RESEARCH QUESTIONS

This survey specifically addresses the following research questions:

TABLE 1. Evolution of LLM evaluation: key milestones and paradigm shifts.

Year	Milestone	Type	Impact
2002	BLEU metric	Metric	Automated MT evaluation standard
2004	ROUGE metric	Metric	Summarization evaluation automated
2018	GLUE benchmark	Suite	Unified NLU evaluation
2019	SuperGLUE	Suite	Response to GLUE saturation
2020	GPT-3 release	Model	Few-shot paradigm shift
2021	MMLU introduced	Benchmark	Expert-level knowledge testing
2021	HumanEval	Benchmark	Code generation standard
2022	BIG-Bench	Suite	204 tasks, emergent abilities
2022	HELM framework	Framework	Holistic multi-metric evaluation
2023	Chatbot Arena	Platform	Live human preference evaluation
2023	SWE-bench	Benchmark	Real-world software tasks
2024-25	Contamination crisis	Challenge	Widespread benchmark leakage

- **RQ1:** What are the fundamental trade-offs between different evaluation paradigms (scalability, cost, reproducibility, authenticity)?
- **RQ2:** How can the lifecycle of evaluation benchmarks be systematically managed to maintain their validity?
- **RQ3:** What is the empirical evidence for the gap between benchmark performance and real-world capability?
- **RQ4:** How can the field transition from leaderboard-driven evaluation to scientifically rigorous assessment?

III. SURVEY METHODOLOGY

A. LITERATURE COLLECTION AND SCOPE

This survey synthesizes research from multiple sources to provide a comprehensive analysis of LLM evaluation:

1) INCLUSION CRITERIA

Papers and benchmarks were included based on the following criteria:

- **Temporal Scope:** Primary focus on publications from 2018-2025, coinciding with the transformer revolution and subsequent LLM development
- **Source Types:** Peer-reviewed conference papers (NeurIPS, ICML, ACL, EMNLP), journal articles, technical reports from major AI labs, and documented benchmarks with public leaderboards
- **Relevance:** Direct contribution to LLM evaluation methodology, benchmark development, metric design, or critical analysis of evaluation practices

2) SEARCH STRATEGY

- **Database Search:** Systematic searches in ACL Anthology, arXiv, IEEE Xplore, and Google Scholar using keywords: “LLM evaluation,” “benchmark,” “metric,” “contamination,” “holistic evaluation”
- **Citation Tracking:** Forward and backward citation analysis from seminal papers (e.g., GLUE, SuperGLUE, MMLU, HELM)

- **Benchmark Analysis:** Direct examination of 20+ major benchmarks, including their documentation, leaderboards, and associated papers

B. ANALYSIS FRAMEWORK

1) TAXONOMY DEVELOPMENT

The five-paradigm taxonomy (Section V) was developed through:

- 1) Initial categorization based on evaluation approach (static vs. dynamic, automated vs. human)
- 2) Iterative refinement based on trade-off analysis (Table 4)
- 3) Validation through mapping of existing benchmarks to categories

2) EMPIRICAL DATA VERIFICATION

All quantitative claims underwent rigorous verification:

- **Correlation Data:** Based on systematic reviews (e.g., Reiter 2018 for BLEU [41])
- **Benchmark Status:** Verified through official benchmark websites and recent literature
- **Performance Gaps:** Documented through empirical studies comparing benchmark and real-world performance

C. LIMITATIONS OF SURVEY SCOPE

This survey acknowledges several scope limitations:

- **Language Bias:** Focus on English-language publications and benchmarks
- **Modality Focus:** Primary emphasis on text-based evaluation with limited multimodal coverage
- **Temporal Cutoff:** Analysis current as of end of 2025; rapid field evolution may introduce new developments

IV. THE EVOLUTION OF LLM EVALUATION: A THREE-ACT NARRATIVE

The history of evaluating language models is not a linear progression but a series of punctuated equilibria. For long

periods, the field coalesces around a set of metrics and benchmarks that define progress. These periods are inevitably disrupted by a new generation of models whose capabilities render the old standards insufficient, prompting the community to develop a new evaluation paradigm. This section outlines this evolution in three acts, from the foundational metrics of classical NLP to the fragmented and complex landscape of today.

A. ACT I: THE PRE-LLM ERA (TASK-SPECIFIC METRICS)

1) FOUNDATIONS IN LEXICAL OVERLAP

The early days of automated evaluation for language generation were dominated by metrics based on a simple, intuitive principle: a good machine-generated text should share many of the same words and phrases as a high-quality human-written reference. This principle gave rise to two of the most influential and enduring metrics in the history of NLP.

BLEU (Bilingual Evaluation Understudy), introduced by Papineni et al. (2002), became the de facto standard for machine translation (MT) [38]. Its core mechanism is *modified n-gram precision*. It calculates the proportion of n-grams (contiguous sequences of n words) in the candidate translation that also appear in a set of human reference translations [41]. To prevent models from gaining high precision by simply repeating common words, the count for each candidate n-gram is “clipped” to the maximum number of times it appears in any single reference translation [38]. Furthermore, BLEU incorporates a *brevity penalty* that penalizes translations that are shorter than the reference texts, discouraging models from producing terse but incomplete outputs [38]. The final BLEU score is a geometric mean of the modified n-gram precisions for $n=1$ to 4, multiplied by the brevity penalty. Its formula is typically expressed as:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right) \quad (1)$$

where BP is the brevity penalty, w_n are weights (typically uniform $1/N$), and p_n is the modified n-gram precision for n-grams of length n .

ROUGE (Recall-Oriented Understudy for Gisting Evaluation), developed by Lin, was designed for evaluating automatic text summarization [31]. As its name suggests, ROUGE is fundamentally recall-oriented, measuring how many of the n-grams in the human reference summaries are captured by the model-generated summary [31]. This contrasts with BLEU’s precision focus. ROUGE is a suite of metrics, with the most common being:

- **ROUGE-N**: Measures the overlap of n-grams (e.g., ROUGE-1 for unigrams, ROUGE-2 for bigrams) [41].
- **ROUGE-L**: Measures the longest common subsequence (LCS), which captures sentence-level structural similarity without requiring consecutive matches [31].

The formula for ROUGE-N recall is:

$$\text{ROUGE-N} = \frac{\sum_{S \in \{\text{RefSummaries}\}} \sum_{\text{gram}_n \in S} \text{Count}_{\text{match}}(\text{gram}_n)}{\sum_{S \in \{\text{RefSummaries}\}} \sum_{\text{gram}_n \in S} \text{Count}(\text{gram}_n)} \quad (2)$$

Despite their widespread adoption, these lexical-overlap metrics have well-documented limitations. Their primary weakness is “semantic blindness”; they cannot recognize synonyms, paraphrases, or semantically equivalent but lexically different phrases [41]. A generated sentence like “The feline sat on the rug” would receive a low score against the reference “The cat was on the mat,” despite conveying nearly identical meaning. Consequently, their correlation with human judgments of quality can be inconsistent and highly dependent on the specific task and dataset [6]. These metrics measure surface-level textual similarity, not deep understanding.

2) THE RISE OF BENCHMARK SUITES

As NLP models grew more sophisticated with the advent of pre-training techniques, the community recognized the need for evaluation that went beyond single tasks. This led to the creation of benchmark suites that aggregated multiple datasets to provide a more general measure of language understanding. As summarized in Table 2, empirical studies confirm that classical automated metrics such as BLEU and ROUGE exhibit highly variable correlations with human judgments, with reasonable system-level reliability in some settings but consistently weak performance at the segment level and across many generation tasks.

The **General Language Understanding Evaluation (GLUE)** benchmark, introduced by Wang et al. (2018), was a landmark in this effort [46]. GLUE was not a new dataset but a curated collection of nine existing NLU tasks, including sentiment analysis (SST-2), textual entailment (RTE, MNLI), and sentence similarity (STS-B) [46]. Its central goal was to drive research toward general and robust NLU systems by favoring models that could transfer knowledge across this diverse set of tasks [46]. By providing a public leaderboard and a single aggregate score, GLUE created a powerful incentive structure that focused a significant portion of the NLP research community on a common goal. This focus directly contributed to the development and popularization of powerful Transformer-based models, such as BERT, which achieved state-of-the-art performance on the benchmark shortly after its release [45]. However, GLUE’s success was also its undoing; the rapid progress of models meant its limited task formats and difficulty level were quickly surpassed, revealing its lack of long-term headroom [45].

B. ACT II: THE TRANSFORMER REVOLUTION AND BENCHMARK SATURATION

1) HOW FOUNDATIONAL MODELS BROKE EXISTING BENCHMARKS

The introduction of large-scale Transformer models, particularly OpenAI’s GPT series, marked a paradigm shift.

TABLE 2. Correlation of automated metrics with human judgment: a synthesis of empirical findings.

Metric	Focus	Reported Correlation Range	Context and Caveats
BLEU	N-gram Precision	0.26-0.40 (System-level)	Correlation is highly variable and context-dependent. Considered reasonable only for system-level MT evaluation. Correlations for other tasks (e.g., NLG) and at the text/segment level are consistently poor [41].
ROUGE-N	N-gram Recall	Varies (High to Low)	Correlation is domain-specific. High for news summarization in DUC/TAC evaluations, but reported as low for meeting summarization unless domain-specific features are incorporated.
ROUGE-L	LCS	Varies (High to Low)	Similar to ROUGE-N, correlation is highly dependent on domain and evaluation setup. Captures sentence-level structure better than n-gram overlap.
METEOR	Precision & Recall (Harmonic Mean)	0.33-0.40 (Sentence-level) up to 0.96 (System-level)	Designed to improve upon BLEU's semantic blindness. A critical distinction exists between modest sentence-level correlation and very high system-level correlation.
BERTScore	Contextual Embedding Similarity	0.48-0.65+	Represents a paradigm shift from lexical overlap to semantic similarity. Consistently shows a stronger correlation with human judgments than prior metrics across various tasks [53].

Unlike previous models that were typically pre-trained and then fine-tuned for specific downstream tasks, models like GPT-2 and GPT-3 demonstrated powerful *in-context learning* capabilities [8]. By providing a few examples within the prompt itself, these models could perform a vast range of tasks they had never been explicitly trained for, from translation to coding to answering trivia questions [40].

This emergent versatility fundamentally broke the existing evaluation paradigm. Benchmarks like GLUE were designed to test a model's ability to be fine-tuned for a handful of specific tasks. They were not designed to measure the broad, open-ended capabilities of a single, massive model. As a result, these benchmarks were rapidly "saturated." Within a year of GLUE's release, top models began to surpass the performance of non-expert humans, rendering the benchmark less useful as a tool for differentiating model capabilities or measuring further progress [45]. This phenomenon, where a benchmark is solved so quickly that it becomes obsolete, highlights the unsustainable nature of static evaluation in an era of exponential model improvement. The question shifted from "Can a model be trained to do X?" to "What can't this single giant model do?"

2) THE RESPONSE: SUPERGLUE, A "STICKIER" BENCHMARK

The research community's direct response to the saturation of GLUE was the creation of **SuperGLUE** [45]. The design of SuperGLUE was a deliberate and thoughtful effort to create a "stickier" benchmark, one that would be more resistant to rapid saturation. It improved upon GLUE in several key ways:

- **More Challenging Tasks:** SuperGLUE retained the most difficult tasks from GLUE and introduced new, more complex ones, selected through an open call for proposals to the research community [45].
- **More Diverse Task Formats:** It moved beyond GLUE's simple sentence-pair classification format to include tasks like coreference resolution and complex question

answering, which require reasoning over longer contexts [45].

- **Comprehensive Human Baselines:** The creators established strong human performance baselines for all tasks, ensuring that a substantial gap existed between the state-of-the-art models at the time and human-level performance, thus providing ample headroom for future research [45].

SuperGLUE represented a more mature approach to benchmark design, consciously anticipating future model improvements. However, even this more rigorous benchmark eventually met the same fate as its predecessor, as continued scaling of language models ultimately conquered its challenges, signaling that the entire paradigm of a fixed suite of NLU tasks was becoming insufficient. Table 3 provides a comparative view of several influential LLM benchmarks, illustrating their number of tasks, time to saturation or human parity, and current status, and making explicit how quickly many once-definitive benchmarks have become obsolete.

C. ACT III: THE CURRENT LANDSCAPE OF PROLIFERATION AND FRAGMENTATION

The saturation of even "sticky" benchmarks, such as SuperGLUE, ushered in the current era, defined by two parallel trends: the creation of massive, broad-coverage benchmarks and an explosion of highly specialized, domain-specific evaluations.

1) THE EMERGENCE OF MASSIVE, MULTITASK BENCHMARKS

Recognizing that a handful of NLU tasks could not capture the full spectrum of an LLM's knowledge or reasoning abilities, researchers began developing benchmarks of unprecedented scale and scope.

- **MMLU (Measuring Massive Multitask Language Understanding):** Introduced by Hendrycks et al.

TABLE 3. Lifecycle and status of major LLM benchmarks.

Benchmark	Year	Tasks	Format	Sat./Parity	Current Status & Key Caveats
GLUE	2018	9	Mixed	12 mo.	Superseded. Saturated within a year of release as models approached or exceeded the reported human baseline, motivating the creation of SuperGLUE as a harder successor [45].
SuperGLUE	2019	8	Mixed	24 mo.	Saturated. Designed as a "stickier" successor to GLUE, it was also saturated as models surpassed the human baseline within two years.
MMLU	2021	57	MC	Ongoing	Active. Remains a challenging benchmark for broad, expert-level knowledge, though top models are approaching the human expert baseline [22].
HumanEval	2021	164	Code	18 mo.	Contamination risk. HumanEval has become a prominent target of contamination analyses, and benchmark performance may be inflated for models exposed to overlapping training data; results should therefore be interpreted cautiously [19].
GSM8K	2021	8,792	Math	24 mo.	Saturated & Noisy. Top models achieve >90% accuracy, but the performance plateau is partly an artifact of significant label noise in the test set. Cleaned versions show different model rankings.

(2021), MMLU represented a significant leap in difficulty and breadth [22]. It consists of over 15,000 multiple-choice questions covering 57 subjects, from elementary mathematics and US history to professional law and medicine [22]. At its release, even the formidable GPT-3 achieved only 43.9% accuracy, far below the estimated human expert level of 89.8%, reestablishing a challenging frontier for the field [22].

- **BIG-Bench (Beyond the Imitation Game Benchmark):** An enormous collaborative effort involving 450 authors, BIG-Bench introduced 204 diverse tasks designed to probe capabilities beyond the scope of existing models [44]. Its tasks span linguistics, commonsense reasoning, math, and even esoteric subjects, with a focus on identifying "emergent abilities" that appear only at a massive scale [44].
- **HELM (Holistic Evaluation of Language Models):** Developed by Liang et al. at Stanford CRFM, HELM introduced a crucial shift in philosophy [30]. Instead of focusing exclusively on performance, HELM advocates a "holistic" approach that evaluates models across seven key metrics: accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency [30]. By standardizing evaluation across 30 major LLMs on 42 scenarios, HELM aims to bring much-needed transparency and comparability to the field, exposing the trade-offs between different model properties [30].

2) THE EXPLOSION OF DOMAIN-SPECIFIC EVALUATION

As general-purpose LLMs demonstrated their potential, specialists in various fields began creating benchmarks to test their utility and safety for specific, high-stakes applications. This has led to a Cambrian explosion of domain-specific evaluations:

- **Code and Software Engineering:** Benchmarks like **HumanEval** [10] and **MBPP** (Mostly Basic Python Programming) test the ability to generate correct code snippets from docstrings. More recently, benchmarks like **SWE-bench** [26] have pushed the frontier toward

more realistic, agentic tasks, requiring models to resolve actual GitHub issues within a full repository context.

- **Medicine and Healthcare:** To assess clinical knowledge and safety, benchmarks like **MedQA** (USMLE-style questions), **PubMedQA**, and the safety-focused **Med-HALT** have become standard. Following the holistic paradigm, **MedHELM** adapts the HELM framework to evaluate models on real-world clinical tasks defined by medical experts [5].
- **Multilingual Capabilities:** To move beyond the English-centric focus of most evaluations, benchmarks like **FLORES-101/200** [20] provide high-quality, human-translated datasets for low-resource machine translation, while **XQuAD** [3] and **MGSM** [43] test cross-lingual question answering and mathematical reasoning, respectively.

3) THE PARADOX OF CHOICE

This Proliferation has created a new challenge: the paradox of choice. With hundreds of benchmarks available, it is unclear which are truly meaningful or reliable. Different research labs often report scores on different subsets of benchmarks, making direct comparison difficult. Worse, even for a single, widely adopted benchmark like MMLU, inconsistent evaluation protocols, such as different prompting strategies or the number of few-shot examples, can lead to significantly different scores for the same model, eroding trust in reported results and making it difficult to track genuine progress [28]. The field has evolved from a unified yet limited evaluation landscape to a rich yet fragmented one, creating a pressing need for standardization and a more scientific approach to measurement.

The entire history of evaluation can be seen as a reactive cycle. Each new generation of models exposes the limitations of the previous generation's evaluation tools, forcing the community to build a new benchmark that is more difficult or more comprehensive. This cat-and-mouse game, while driving progress, is inherently inefficient. It keeps the focus on task performance ("what" models can do) rather than

on understanding the underlying capabilities (“how” and “why” they work). A mature science of evaluation must break this cycle, moving toward proactive design based on principles from cognitive science and metrology, a theme we will develop throughout this paper.

V. A TAXONOMY OF EVALUATION METHODOLOGIES

The rapid growth of LLMs has been matched by a corresponding expansion of methods to evaluate them. The landscape is now a complex ecosystem of static datasets, interactive environments, human-powered platforms, and even AI-driven judges. To navigate this complexity, a systematic classification is required. This section introduces a comprehensive taxonomy of LLM evaluation methodologies, visualized in Figure 2. We organize the fragmented landscape into five distinct paradigms to enable principled comparison. This framework provides a structured approach to understanding the strengths, weaknesses, and underlying trade-offs of different approaches, enabling researchers and practitioners to make informed choices about how to measure model capabilities.

A. STATIC BENCHMARKS: THE TRADITIONAL PARADIGM

Static benchmarks represent the classical and still most common paradigm for LLM evaluation. They consist of fixed, offline datasets containing a set of inputs (e.g., questions, prompts) and corresponding ground-truth outputs (e.g., correct answers, reference texts). Performance is typically measured by automated metrics that compare a model’s generated output against the ground truth.

1) FOUNDATIONAL CAPABILITIES

A core set of static benchmarks has become the de facto standard for measuring the general capabilities of new LLMs. These include:

- **MMLU (Massive Multitask Language Understanding):** A challenging multiple-choice benchmark designed to measure broad knowledge and problem-solving ability across 57 subjects in STEM, humanities, and other areas [22].
- **HellaSwag:** A commonsense natural language inference task that requires choosing the most plausible continuation of a given text context. Its incorrect options are adversarially generated to be challenging for models but trivial for humans [52].
- **ARC (AI2 Reasoning Challenge):** A collection of grade-school science questions designed to test complex reasoning. It includes an “Easy Set” and a “Challenge Set” containing questions that simple retrieval and co-occurrence algorithms fail to answer [13].
- **GSM8K (Grade School Math):** A dataset of grade-school math word problems that require multiple steps of arithmetic reasoning to solve. It tests a model’s ability to parse natural language, identify the necessary calculations, and execute them correctly [14].

2) SPECIALIZED DOMAINS

To assess performance on more specific, high-value tasks, numerous domain-specific benchmarks have been developed:

- **Code Generation: HumanEval** is a widely used benchmark consisting of 164 hand-written Python programming problems. Models are evaluated based on the functional correctness of their generated code, as determined by unit tests [10].
- **Mathematical Reasoning: MathQA** provides a large-scale dataset of math word problems annotated with operation-based formalisms, aiming for more interpretable evaluation [2].
- **Scientific Reasoning: ScienceQA** is a multimodal benchmark featuring science questions that may include text, images, or both, along with detailed explanations and lectures that serve as a chain-of-thought [34].

3) MULTILINGUAL EVALUATION

To counter the English-centric bias of most benchmarks, several have been created to evaluate performance across many languages:

- **FLORES-101/200:** A high-quality benchmark for low-resource machine translation, providing professionally translated sentences that are fully aligned across over 100 languages [20].
- **XQuAD:** A cross-lingual question-answering dataset based on SQuAD v1.1, with paragraphs and questions professionally translated into ten other languages to facilitate zero-shot cross-lingual transfer evaluation [3].
- **MGSM (Multilingual Grade School Math):** A multilingual version of a subset of GSM8K, translated into ten typologically diverse languages to test multilingual mathematical reasoning [43].

4) CRITICAL ANALYSIS

The primary **strengths** of static benchmarks are their high reproducibility and low computational cost for evaluation (once the dataset is created). They provide a standardized basis for comparing models, which has been instrumental in driving progress. However, their **weaknesses** are becoming increasingly apparent. They are highly vulnerable to **data contamination**, where test set examples are inadvertently included in the massive, uncuration datasets used to train LLMs, leading to artificially inflated scores [12]. This encourages overfitting to the specific artifacts of a benchmark rather than fostering true generalization. As models advance, these benchmarks can quickly become outdated and fail to differentiate between state-of-the-art models [45].

B. DYNAMIC AND ADAPTIVE EVALUATION

To overcome the limitations of static benchmarks, a new paradigm of dynamic and adaptive evaluation has emerged. In this approach, the evaluation is not a one-time event on a fixed dataset but a continuous process where the test set evolves, often in response to the very models being evaluated.

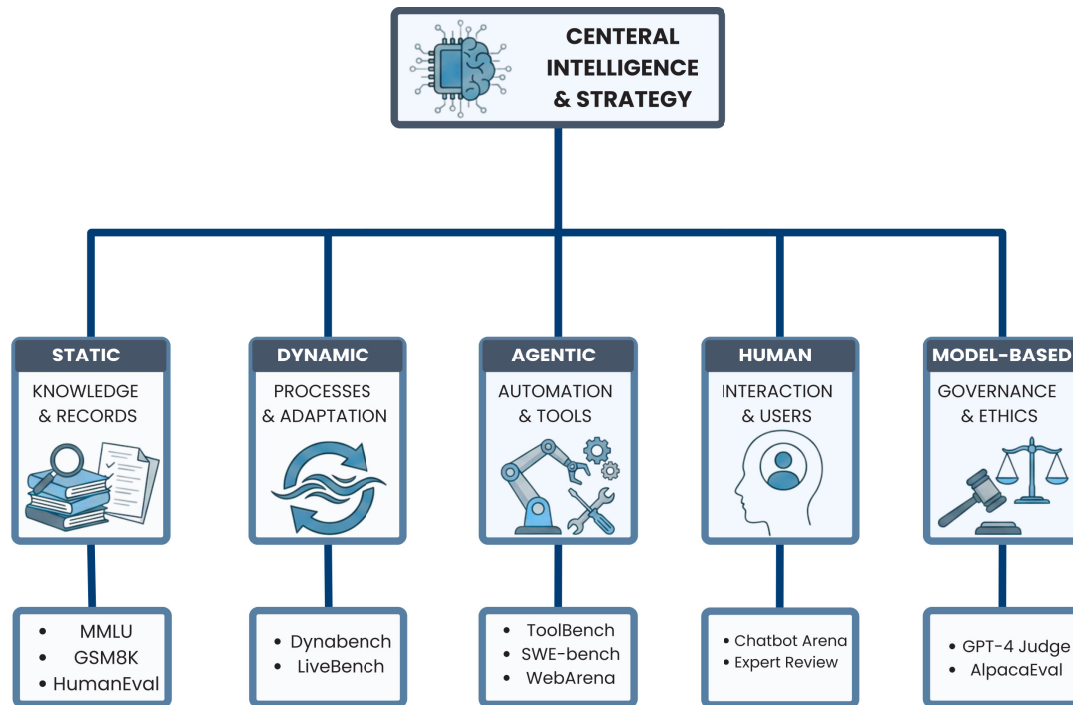


FIGURE 2. A taxonomy of evaluation methodologies. This hierarchical view organizes current evaluation strategies into five distinct paradigms, ranging from static datasets to dynamic, agentic, and model-based approaches.

- **Living Benchmarks:** These are benchmarks designed to be continuously updated with new, challenging data to stay ahead of model capabilities and mitigate contamination. For example, LiveBench releases new questions regularly and focuses on tasks with verifiable, objective answers to avoid the need for subjective LLM judges.
- **Adversarial Evaluation (Red-Teaming):** This involves a human-and-model-in-the-loop process where human annotators or automated systems actively search for inputs that cause a model to fail or exhibit unsafe behavior [27]. The goal is to discover “unknown unknowns”, the weaknesses that were not anticipated during development.
- **Case Study: Dynabench:** Dynabench is a research platform that operationalizes this dynamic approach [27]. It hosts NLP tasks where human annotators compete against a target model. The annotators’ goal is to create examples that fool the model but are correctly classified by other humans. These “model-in-the-loop” generated examples are then collected into new datasets that are adversarially challenging by construction. This creates a “moving target” for models, ensuring the benchmark does not saturate and continuously surfaces model weaknesses [27].

C. INTERACTIVE AND AGENTIC EVALUATION

The most significant shift in evaluation is the move from assessing passive text generation to evaluating active, goal-oriented agents. These evaluations measure a model’s ability

to perform complex, multi-step tasks by interacting with an environment, using tools, and maintaining state over time.

- **Multi-turn Dialogue:** Evaluating conversational agents requires assessing not just the quality of a single response but the coherence, consistency, and helpfulness of an entire dialogue. **MT-Bench** is a key benchmark in this area, featuring 80 challenging multi-turn questions that test instruction-following and reasoning [4]. **Chatbot Arena** takes a different approach, using crowd-sourced, pairwise comparisons of anonymous models in live conversations to generate a dynamic Elo rating-based leaderboard [55].
- **Tool Use and Function Calling:** Modern LLMs can interact with external tools like APIs and databases. Benchmarks such as **ToolBench** [39] and **API-Bank** [29] evaluate a model’s ability to correctly identify the need for a tool, select the right API from an extensive library, formulate the correct call with the proper parameters, and appropriately use the returned information.
- **Agent Benchmarks:** The most advanced form of interactive evaluation places LLM-powered agents in complex environments to complete high-level goals. **AgentBench** provides a suite of eight diverse environments, from operating a computer to online shopping [33]. **WebArena** provides a realistic and reproducible set of web-based tasks, evaluating agents on their ability to perform actions such as booking flights or managing content on a wiki [56]. In the software engineering domain, **SWE-bench** tasks agents with

resolving real-world GitHub issues in full-scale code repositories [26].

The primary challenge in this paradigm is the inherent non-determinism and statefulness of agentic systems, which makes evaluation complex, costly, and difficult to reproduce reliably.

D. HUMAN-IN-THE-LOOP EVALUATION

Despite the push for automation, human evaluation remains the gold standard for assessing qualities that are nuanced, subjective, or context-dependent. It is indispensable for measuring aspects like creativity, helpfulness, and safety, which are poorly captured by automated metrics.

- **Direct Assessment:** Human raters, either crowdworkers or domain experts, score model outputs on one or more criteria (e.g., fluency, factuality, harmlessness) using a predefined scale, such as a Likert scale from 1 to 5.
- **Pairwise Comparison:** Instead of providing an absolute score, raters are shown two anonymous model outputs for the same prompt and asked to choose which one is better. This relative judgment is often easier for humans to make consistently and forms the basis of platforms like Chatbot Arena [55].
- **Expert vs. Crowdsourced Evaluation:** For specialized domains, evaluation relies on experts (e.g., doctors for medical tasks, lawyers for legal reasoning) to ensure accuracy and nuance [5]. For general tasks, crowdsourcing platforms like Amazon Mechanical Turk or Surge AI are often used to achieve scalability.

The main challenges of human evaluation are its high cost, slow turnaround time, and subjectivity. **Inter-rater agreement**, the degree to which different human raters agree, can often be low, especially for complex or subjective tasks. This raises questions about the reliability of human evaluation itself, with recent studies highlighting a “reproducibility crisis” in which as few as 5% of published human evaluations provide sufficient information to be repeatable [6].

E. MODEL-BASED EVALUATION (LLM-AS-A-JUDGE)

To address the scalability limitations of human evaluation, the “LLM-as-a-Judge” paradigm has gained significant traction [37]. This approach uses a powerful, state-of-the-art LLM (e.g., GPT-4o) as a proxy for a human evaluator.

- **The Paradigm:** The judge LLM is given a prompt that defines the evaluation criteria, along with the model output(s) to be assessed. It can perform both single-output scoring (assigning a score on a scale) and pairwise comparison (choosing the better of two responses) [9].
- **Validation:** This method has been validated by showing that the judgments of top-tier LLMs correlate highly with human judgments, often achieving over 80% agreement, which is comparable to inter-human agreement rates [9]. This makes it a scalable and cost-effective alternative for many tasks.

- **Risks and Critiques:** The primary risk of this approach is that the judge LLM is not an objective oracle; it has its own inherent biases. Documented biases include **position bias** (favoring the first response), **verbosity bias** (favoring longer responses), and **self-preference bias** (favoring outputs that resemble its own style or are from the same model family) [37]. Over-reliance on a single model family as the judge (e.g., GPT-4) risks **bias amplification**, where the stylistic preferences of that family become the de facto standard for the entire field, leading to a problematic **circular evaluation** loop.

This taxonomy reveals a fundamental trade-off in LLM evaluation, spanning a spectrum from highly scalable and reproducible but potentially less authentic methods (static benchmarks) to highly authentic but unscalable and subjective methods (human evaluation). The most innovative recent approaches, such as agentic evaluation and LLM-as-a-Judge, can be understood as attempts to find a more optimal point on this spectrum, balancing the competing demands of rigor, relevance, and scalability. A mature evaluation strategy must therefore consist of a portfolio of methods drawn from across this taxonomy, selected based on the specific goals and constraints of the evaluation. The trade-offs discussed in this taxonomy are summarized in Table 4, which contrasts the five evaluation paradigms along dimensions such as scalability, cost, reproducibility, authenticity, and contamination risk.

VI. THE METROLOGICAL SCIENCE OF LLM EVALUATION

To move beyond the current evaluation crisis, the field must adopt a more rigorous and principled approach, treating evaluation as a form of metrology, the science of measurement. This involves not only designing better measurement instruments (benchmarks) but also understanding their properties, managing their lifecycle, and applying sound statistical principles to their results. This section introduces two core components of this new science: the Benchmark Lifecycle Framework, a novel paradigm for managing evaluation instruments, and a set of principles for ensuring statistical severity in reporting and interpreting results.

A. THE BENCHMARK LIFECYCLE FRAMEWORK

Benchmarks are not timeless truths; they are scientific tools with a finite lifespan. Their utility degrades over time due to model saturation, data contamination, and shifts in research priorities. We propose the **Benchmark Lifecycle Framework** as a systematic process for creating, deploying, maintaining, and retiring benchmarks. This framework, illustrated in Figure 3, consists of five key phases. Table 5 offers a concise overview of the Benchmark Lifecycle Framework, specifying the key activities associated with each phase from initial design and validation through deployment, monitoring, and eventual retirement.

TABLE 4. Evaluation methodology matrix: a comparative analysis of key trade-offs.

Property	Static Benchmarks	Dynamic/ Adaptive	Interactive/ Agentic	Human-in-the-Loop	Model-Based (LLM-as-Judge)
Scalability	High	Medium	Low	Very Low	Very High
Cost (per run)	Low	Medium	High	Very High	Low
Reproducibility	Very High	Medium	Low	Low	Medium
Authenticity	Low	Medium	High	Very High	High (but biased)
Contamination Risk	Very High	Low	Medium	N/A	N/A
Expertise Required	Low	Medium-High	High	Varies	Low-Medium

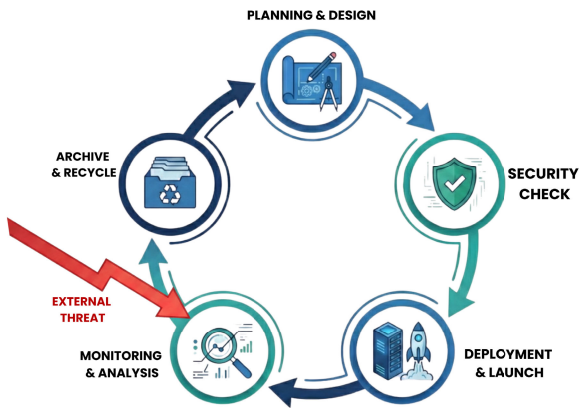


FIGURE 3. The benchmark lifecycle framework. A proposed circular model for managing evaluation instruments, emphasizing that benchmarks have a finite lifespan and require continuous monitoring for data contamination before eventual retirement.

1) DESIGN PHASE

The creation of a benchmark must be a principled engineering process, not an ad-hoc collection of data.

- **Task Selection and Formulation:** The process begins with a clear definition of the capability to be measured. This should be grounded in real-world use cases or well-defined cognitive science principles. The task should be formulated to minimize ambiguity and ensure that it genuinely tests the intended capability.
- **Data Collection and Validation:** Data must be sourced and curated with care to ensure quality, diversity, and representativeness. This includes rigorous validation to remove errors, inconsistencies, and potential biases. Ethical considerations, such as data privacy and the representation of different demographic groups, must be addressed from the outset.
- **Difficulty Calibration:** A benchmark is most useful when it is challenging but not impossible for current state-of-the-art models. The design phase should explicitly calibrate difficulty, for instance, by including tasks that require multi-step reasoning or by using adversarial filtering techniques to generate challenging distractors, as seen in HellaSwag [52].

2) VALIDATION PHASE

Before a benchmark is widely adopted, it must go through rigorous validation to ensure it is a reliable and meaningful instrument.

- **Statistical Power Analysis:** The benchmark must contain a sufficient number of test instances to reliably detect statistically significant differences in performance between models. Small test sets can produce noisy results and lead to spurious conclusions about which model is superior.
- **Cross-Validation with Human Performance:** Establishing a strong human baseline is critical. This not only sets a clear goal for AI systems but also helps validate that the task is well-defined and solvable. A large gap between model performance and human performance indicates that the benchmark has sufficient “headroom” for future progress [45].
- **Robustness Testing:** The benchmark should be tested for brittleness. Small, semantically irrelevant perturbations to the test inputs should not significantly impact model performance. This helps ensure the benchmark measures robust understanding rather than superficial pattern matching.

3) DEPLOYMENT PHASE

The release and maintenance of a benchmark require careful management to ensure its long-term value.

- **Standardized Evaluation Protocols:** To ensure that results are comparable across different studies, benchmark creators must provide clear, unambiguous, and standardized protocols for evaluation. This includes specifying prompt formats, the number of few-shot examples, and the exact code for calculating metrics. The fragmentation of evaluation protocols for MMLU is a cautionary tale of why this is essential [28].
- **Version Control and Reproducibility:** Benchmarks should be treated like software, with clear versioning. Any changes to the dataset or evaluation protocol should be documented and assigned a new version number, allowing researchers to ensure they are comparing results generated under identical conditions.

4) CONTAMINATION DETECTION PHASE

Given the inevitability of contamination in the era of web-scale training data, active and continuous monitoring is a crucial part of the lifecycle.

- **Proactive Monitoring:** Benchmark maintainers and the community at large should proactively run contamination detection methods on new models as they are released.

TABLE 5. Benchmark lifecycle phases and key activities.

Phase	Key Activities
Design	Task selection and formulation, data collection and validation, difficulty calibration
Validation	Statistical power analysis, establishing human baselines, robustness testing
Deployment	Protocol standardization, version control, documentation
Monitoring	Proactive contamination detection, performance tracking, anomaly detection
Retirement	Saturation assessment, transition planning, archival

- **Detection Methods:** A variety of methods can be employed, from simple n-gram overlap checks to more sophisticated techniques like the **Data Contamination Quiz (DCQ)**, which tests a model’s ability to distinguish an original test instance from slightly perturbed versions [19]. Anomalous performance patterns, in which a model performs exceptionally well on a specific benchmark despite its general capabilities, can also be a strong signal of contamination.

5) RETIREMENT CRITERIA

All benchmarks eventually become obsolete. A mature evaluation ecosystem requires a clear process for retiring them.

- **Retirement Triggers:** Clear criteria for retirement should be established. These can include: (1) **Saturation**, where top models consistently achieve near-perfect or superhuman performance; (2) **Widespread Contamination**, where detection methods reveal that the benchmark is present in the training sets of most major models; or (3) **Irrelevance**, where the task measured by the benchmark is no longer a key research challenge.
- **Transition Strategies:** The community, led by standards bodies and conference organizers, should develop clear strategies to transition away from retired benchmarks to next-generation evaluation tools, preventing the perpetuation of outdated and uninformative metrics.

B. STATISTICAL RIGOR IN EVALUATION

The prevailing culture of reporting single-point scores on leaderboards obscures uncertainty and often leads to misleading conclusions. A metrological approach demands greater statistical rigor.

- **Proper Hypothesis Testing:** Comparing models should be framed as a statistical hypothesis test. Instead of simply observing that Model A’s score of 85.4 is higher than Model B’s score of 85.2, researchers should test the null hypothesis that the models’ performance is not statistically different and report the corresponding p-values.
- **Confidence Intervals and Effect Sizes:** A single accuracy score is a point estimate from a sample (the test set). It should always be accompanied by a confidence interval (e.g., using bootstrapping) to quantify the uncertainty in the measurement. Furthermore, reporting the effect size (e.g., Cohen’s d) indicates the magnitude of the performance difference, distinguishing between statistically significant but practically meaningless improvements and substantial advances.

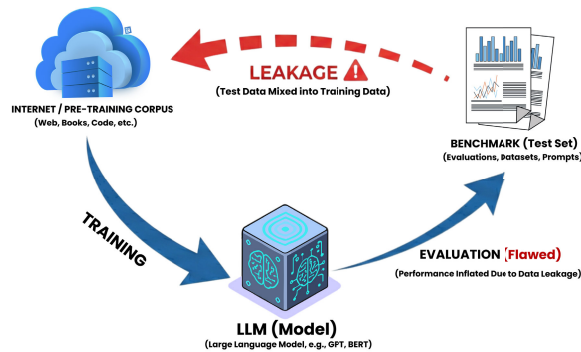


FIGURE 4. The mechanism of data contamination. Because benchmarks are often uploaded to the internet, they are inadvertently scraped into training corpora. This allows models to “memorize” the test set, creating a feedback loop where evaluation scores no longer reflect genuine reasoning.

- **Multiple Comparison Corrections:** A common pitfall is making spurious “state-of-the-art” (SOTA) claims by comparing dozens of models across dozens of benchmarks without correcting for multiple comparisons. When many tests are performed, some will appear significant by random chance alone. Methods like the Bonferroni or Benjamini-Hochberg correction should be applied to control the family-wise error rate or false discovery rate, leading to more credible SOTA claims.
- **Sample Size Considerations:** The size of the evaluation set directly impacts the statistical power of the evaluation, the ability to detect a true performance difference. Researchers must consider the trade-off between the cost of evaluation and the desired level of statistical confidence.

C. THE CONTAMINATION CRISIS

Data contamination is arguably the most urgent and corrosive threat to the validity of LLM evaluation today. The practice of training models on vast, unfiltered web scrapes makes it almost certain that data from popular benchmarks will be included in the training set. Figure 4 demonstrates this

feedback loop, showing how test data leakage invalidates subsequent evaluations.

1) TYPES OF CONTAMINATION

Contamination can manifest in several forms:

- **Direct Inclusion:** The most blatant form, where test set examples are directly included in the training data [19].
- **Paraphrased or Augmented Versions:** Semantically identical but syntactically varied versions of test examples are present in the training data.
- **Semantic Overlap (Leakage):** The training data contains information that is not identical to the test examples, but makes solving them insignificant. For example, a training corpus might contain a detailed explanation of a specific logical puzzle that later appears in a reasoning benchmark [12].

2) DETECTION METHODS

Detecting contamination is a major challenge, especially for closed models where the training data is inaccessible.

- **Membership Inference Attacks:** These methods attempt to determine if a specific data point was in the training set, often by analyzing the model's output probabilities or confidence levels.
- **Anomalous Performance Patterns:** A model showing surprisingly high performance on a specific, obscure benchmark relative to its performance on other, more general benchmarks can be a strong indicator of contamination.
- **Canary Strings and Watermarking:** Proactively embedding unique, identifiable strings ("canaries") into benchmark datasets can make it easier to detect if that data is later ingested by a model.
- **Perturbation-Based Methods:** The Data Contamination Quiz (DCQ) method frames detection as a multiple-choice question where the model must identify the original benchmark instance from among several slightly altered versions. A model that has memorized the instance will be able to identify it with high accuracy [19].

Table 6 compares existing contamination detection methods along practical dimensions such as accuracy, scalability, data and access requirements, and false positive rate, highlighting that no single technique suffices in isolation.

3) CASE STUDIES

The impact of contamination is not theoretical. The GPT-4 technical report itself acknowledged contamination on several benchmarks, including HumanEval and DROP [19]. The surprisingly high performance of some models on benchmarks like MMLU has been questioned on the grounds of potential contamination, as practice questions and answers are widely available online [22]. The stark performance difference of OpenAI's o3 model on the older ARC-AGI-1 versus the newly released ARC-AGI-2 benchmark strongly

suggests that its high performance on the former may be due to contamination rather than genuine reasoning ability [19]. This issue creates a fundamental paradox: the very method used to make models more powerful, scaling up training data from the web, simultaneously corrupts our ability to reliably measure their power. This self-defeating cycle necessitates a move toward dynamic evaluation and a robust lifecycle management framework that treats contamination not as an occasional accident but as an inevitability to be actively managed.

The current approach to LLM evaluation is plagued by several systemic issues, or pathologies, that distort research incentives, create a misleading picture of AI progress, and neglect crucial dimensions of model performance. This section provides a critical analysis of these pitfalls, from the obsessive pursuit of leaderboard rankings to the profound mismatch between what is measured and what truly matters.

D. THE LEADERBOARD OBSESSION

Public leaderboards, which rank models based on their performance on shared benchmarks, have been a powerful engine for progress in NLP [16]. They create a clear, quantifiable, and competitive target that focuses community effort. However, this intense focus has given rise to a "leaderboard obsession" with several pathological consequences.

1) GOODHART'S LAW IN AI

The most significant issue is the manifestation of Goodhart's Law, an adage from economics that states, "When a measure becomes a target, it ceases to be a good measure." In the context of LLMs, when a benchmark score (the measure) becomes the primary goal for a research lab or company (the target), development efforts shift from genuine capability improvement to optimizing for the specific quirks and artifacts of that benchmark. Models become exquisitely tuned to the statistical distribution of the test set, a phenomenon often referred to as "overfitting to the benchmark" or "gaming" the evaluation. The benchmark score, once a proxy for a general capability like "reasoning," becomes a measure of a much narrower skill: "the ability to get a high score on that specific set of questions." This leads to a situation where scores rise, but true, generalizable capability stagnates or even declines.

2) THE SOTA RACE: SHORT-TERM GAINS VS. LONG-TERM PROGRESS

The intense competition to achieve "state-of-the-art" (SOTA) status on prominent leaderboards creates a culture that prioritizes short-term, incremental gains over long-term, foundational research [16]. Research is incentivized to focus on small architectural tweaks or prompt engineering tricks that can eke out another fraction of a percentage point on a benchmark, rather than exploring novel ideas, creating new and more meaningful datasets, or conducting deep analyses of model failures [16]. This "SOTA race" can

TABLE 6. Data contamination detection methods: illustrative comparison and effectiveness.

Method	Accuracy	Scalability	Requirements	False Positive Rate
N-gram Overlap	Low (40-60%)	High	Training data access	High
Membership Inference	Medium (60-75%)	Medium	Closed-box access	Medium
Data Contamination Quiz (DCQ)	High (75-85%)	Medium	Benchmark variations	Low
Anomaly Detection	Medium (65-80%)	High	Baseline comparisons	Medium
Canary Strings	Very High (95-99%)	Low	Proactive design	Very Low
Temporal Analysis	Medium (70-85%)	High	Historical data	Medium

lead to a hill-climbing dynamic in which the community becomes trapped in a local optimum of a particular evaluation paradigm, neglecting other important performance dimensions, such as efficiency, fairness, or robustness [16].

E. THE EVALUATION-CAPABILITY MISMATCH

A core pathology of the current landscape is the growing divergence between what benchmarks measure and the capabilities we actually want from LLMs. High scores on leaderboards are increasingly failing to correlate with reliable performance in real-world applications.

1) SURFACE FORM VS. UNDERSTANDING

Many static benchmarks, particularly those with multiple-choice or exact-match formats, can be solved through sophisticated pattern matching rather than genuine understanding or reasoning. LLMs, as masters of statistical correlation, are exceptionally good at learning the superficial cues and biases present in a dataset [52]. A model might learn, for example, that questions containing the word “not” are more likely to have “false” as the correct answer in a particular textual entailment dataset. This allows it to achieve a high score without ever engaging with the semantic content of the text. This leads to models that appear competent on the benchmark but are breakable and fail when presented with problems that deviate even slightly from the training distribution.

2) THE CORRELATION PROBLEM

The ultimate test of an evaluation is its correlation with real-world utility. The crisis of confidence stems from repeated observations that this correlation is breaking down [12]. There are numerous documented cases of models that achieve top scores on benchmarks like MMLU or medical exams but fail at simple reasoning tasks or provide unsafe medical advice in practice [5]. For example, a recent study found that while many LLMs can pass exams, their performance on a set of real-world clinical tasks revealed significant gaps, highlighting the inadequacy of exam-based evaluation [5]. This mismatch suggests that our current benchmarks are failing as proxies for the complex, multi-faceted, and context-dependent capabilities required for successful deployment. Table 7 aggregates documented cases across medicine, law, coding, mathematics, and

commonsense reasoning where strong benchmark scores fail to translate into reliable real-world performance, making the evaluation–capability mismatch concrete.

F. THE COST AND SCALE CHALLENGE

The scale of modern LLMs and the complexity of their evaluation have introduced significant logistical and economic challenges.

1) COMPUTATIONAL EXPENSE

Evaluating a large-scale LLM across a comprehensive suite of benchmarks is a computationally intensive and expensive process [48]. Each evaluation run can consume a significant amount of GPU hours, contributing to the substantial carbon footprint of AI development. This high cost creates a barrier to entry, concentrating the ability to perform thorough evaluations in the hands of a few large, well-resourced industrial labs. This can stifle independent academic research and third-party auditing, leading to a less transparent and competitive ecosystem.

2) HUMAN EVALUATION BOTTLENECKS

While human evaluation is often considered the gold standard, it is a major bottleneck due to its high cost, slow speed, and scalability limitations. Assembling and training a team of qualified human raters, especially domain experts for specialized tasks, is a significant undertaking. The sheer volume of outputs generated by LLMs makes it impractical to conduct a comprehensive human evaluation of every model version. This bottleneck often leads to evaluation lagging far behind model development, with thorough assessments performed only long after a model has been deployed.

3) THE EVALUATION-DEVELOPMENT GAP

There is a growing gap between the pace of model development and the pace of evaluation science. New models and capabilities are released at a dizzying rate, while developing robust benchmarks is a slow, painstaking process. This means that by the time a new benchmark is created and validated, it may already be on the verge of obsolescence. This gap ensures that evaluation remains in a reactive, catch-up mode, rather than proactively guiding the development of more capable and aligned AI systems.

TABLE 7. Examples of reported gaps between benchmark performance and real-world application.

Domain	Benchmark Task & Reported Score	Real-World Analogue & Outcome	Nature of the Gap
Medical Diagnosis	MedQA (USMLE questions) >90% accuracy	Clinical Practice: Physicians often outperform LLMs; prone to safety-critical errors	Exam scores are poor proxy for applied reasoning skills required in clinical practice. Benchmark lacks construct validity for predicting real-world clinical competence [5].
Legal Reasoning	Bar Exam QA >75% accuracy	Professional Legal Work: High variance in preference rankings by senior partners	Traditional accuracy metrics misaligned with subjective, strategic, and context-dependent qualities legal professionals value.
Code Generation	HumanEval >84% pass@1	Software Engineering: Performance on realistic SWE-bench significantly lower; more high-severity security flaws	Benchmark measures only functional correctness on isolated problems, ignoring maintainability, security, efficiency, and integration requirements.
Math Reasoning	GSM8K >92% accuracy	Robust Problem Solving: Performance brittle, drops with minor changes	Benchmark measures pattern recognition on noisy dataset, not genuine, generalizable mathematical reasoning. High scores don't reflect robust understanding.
Common Sense	HellaSwag >95% accuracy	General Intelligence: Little empirical evidence correlating scores with real-world capability	Benchmark tests specific, adversarially constructed task. Success may reflect advanced statistical artifact detection rather than general cognitive capability.

G. FAIRNESS AND REPRESENTATION

A critical but often overlooked pathology is the significant bias embedded in the evaluation landscape itself. The benchmarks we use to define “good” performance are predominantly created by and for a narrow slice of the global population.

1) GEOGRAPHIC AND CULTURAL BIAS

The vast majority of influential LLM benchmarks are Western-centric and developed in North America or Europe. They reflect the cultural norms, values, and knowledge priorities of these regions. Tasks and questions often contain cultural assumptions that may not be universally applicable, potentially penalizing models that are trained on more diverse global data or that exhibit different cultural perspectives.

2) LANGUAGE INEQUALITY

The dominance of English in LLM evaluation is overwhelming. While multilingual benchmarks like FLORES and XQuAD exist, they are the exception rather than the rule. Most SOTA claims are based on performance on English-language tasks, creating a powerful incentive to allocate resources to English-language models. This perpetuates a cycle in which models for low-resource languages are underdeveloped and under-evaluated, thereby exacerbating the digital language divide.

3) DEMOGRAPHIC BLINDSPOTS

The data used to create benchmarks often lacks demographic diversity, underrepresenting various racial, ethnic, gender, and socioeconomic groups [36]. This means that a model can achieve a high score on a benchmark while still performing poorly or exhibiting harmful biases for users from underrepresented groups. Without evaluations that explicitly measure performance across demographic subgroups, we are effectively blind to these issues of fairness.

Holistic frameworks like HELM, which include fairness as a core metric, are a crucial step in addressing this blind spot, but they remain the exception in a field still largely driven by aggregate accuracy scores [30].

VII. DOMAIN-SPECIFIC EVALUATION LANDSCAPES

While general benchmarks provide a broad measure of capability, the true utility and risk of LLMs are often revealed only when they are evaluated in specific, high-stakes domains. Each domain presents unique evaluation challenges, requiring specialized benchmarks and metrics that capture the nuances of the task. This section surveys the evaluation landscapes in four critical domains: code and software engineering, scientific and mathematical reasoning, medicine and healthcare, and multimodal applications. Table 8 summarizes the current domain-specific benchmark landscape, outlining the dominant benchmarks, their primary focus, and the unique evaluation challenges that arise in each setting.

A. CODE AND SOFTWARE ENGINEERING

The ability of LLMs to understand and generate code has been one of their most impactful applications. Evaluation in this domain has evolved from testing simple function generation to assessing complex, agentic software engineering tasks.

1) FROM FUNCTIONS TO FULL SYSTEMS

Early code benchmarks focused on function-level code generation. **HumanEval** [10] and the similar **MBPP** benchmark provide short programming tasks defined by a function signature and a docstring, and evaluate models based on whether the generated code passes unit tests. These benchmarks measure fundamental coding correctness and algorithmic understanding.

More recently, the field has moved toward more realistic and complex evaluations. **SWE-bench** represents a significant step up in difficulty and realism [26]. Instead

TABLE 8. Domain-specific benchmark landscape.

Domain	Key Benchmarks	Primary Focus	Unique Challenges	Evaluation Type
Code	HumanEval, MBPP, SWE-bench	Function/system generation	Execution safety, equivalence	Functional correctness
Medicine	MedQA, PubMedQA, Med-HALT	Clinical knowledge, safety	Patient safety, regulations	Accuracy + safety
Mathematics	GSM8K, MATH, MiniF2F	Problem solving, proofs	Formal reasoning	Exact match, verification
Science	ScienceQA, ARC, SciQ	Knowledge, reasoning	Multimodal, experimental	Multiple choice, explanation
Multilingual	FLORES, XQuAD, MGSM	Translation, QA, reasoning	Low-resource, cultural	BLEU, F1, exact match

of generating standalone functions, models are tasked with resolving real-world GitHub issues in large, popular open-source Python repositories such as Django [26]. This requires an agentic workflow: the model must understand the issue, navigate the codebase to locate the relevant files, write a patch to fix the bug, and ensure the patch passes the project's test suite. This evaluates a much richer set of skills, including code comprehension, debugging, and tool use within a complex environment.

2) EVALUATION CHALLENGES

Evaluating code generation presents unique challenges:

- **Correctness, Efficiency, and Readability:** Functional correctness, assessed via unit tests, is the primary metric. However, it is not the only measure of code quality. A correct solution may be inefficient, unreadable, or challenging to maintain. Developing automated metrics for these more subjective qualities remains an open research problem.
- **The Execution Problem:** Evaluating code requires executing it, which poses significant security risks. Untrusted model-generated code must be run in a secure, sandboxed environment to prevent malicious actions, such as accessing the file system or making network calls. This adds a layer of engineering complexity to the evaluation process [25].
- **Equivalence:** There are often many correct ways to solve a programming problem. Simple string-matching metrics are insufficient; evaluation must be based on functional equivalence as determined by a robust test suite.

B. SCIENTIFIC AND MATHEMATICAL REASONING

Mathematics and science are key domains for testing the limits of an LLM's logical reasoning and knowledge application capabilities. Benchmarks in this area are designed to distinguish rote memorization from genuine, multi-step deductive reasoning.

1) TESTING FORMAL REASONING

Benchmarks like **GSM8K** [14] and **MATH** test mathematical problem-solving. GSM8K focuses on grade-school level word problems that require a chain of arithmetic operations, while MATH includes more challenging competition-level

math problems. **MathQA** provides a large-scale dataset with annotations that break down solutions into a sequence of formal operations, aiming for more interpretable evaluation [2].

In the sciences, **ScienceQA** is a multimodal benchmark that includes questions from natural, social, and language sciences, often accompanied by images and detailed, step-by-step explanations [34]. These tests not only test knowledge but also the ability to synthesize information from multiple sources and modalities.

2) PROOF VERIFICATION AND GENERATION

A frontier in mathematical reasoning is the evaluation of formal theorem proving. Benchmarks like **MiniF2F** and **ProofNet** test the ability of models to generate or verify formal proofs in languages like Lean or Isabelle. This is a particularly rigorous form of evaluation, as the correctness of a proof can be checked mechanically by a proof assistant, providing an unambiguous ground truth.

3) THE FORMALIZATION CHALLENGE

A primary challenge in this domain is the gap between natural language and formal systems. Many real-world scientific and mathematical problems are expressed in natural language, but their solutions require translation into a formal, symbolic representation that can be manipulated logically. Evaluating a model's ability to perform this crucial formalization step is an active area of research. Furthermore, as with other domains, the risk of contamination is high, as many math problems and their solutions are available online [11].

C. MEDICAL AND HEALTHCARE

The application of LLMs in medicine is one of the highest-stakes domains, where errors can have severe consequences. Evaluation must therefore prioritize safety, reliability, and clinical validity above all else.

1) CLINICAL KNOWLEDGE ASSESSMENT

Many medical benchmarks are based on professional licensing examinations. **MedQA** [5] uses questions from the United States Medical Licensing Examination (USMLE), while other benchmarks like **PubMedQA** test the ability to answer biomedical research questions based on scientific abstracts. While passing these exams is an impressive feat, there is a growing consensus that such multiple-choice tests are poor proxies for real-world clinical competence [5].

2) SAFETY-CRITICAL EVALUATION

The most critical aspect of medical LLM evaluation is safety. Benchmarks like **Med-HALT** (Harmful Answers from Large Language Models in the Medical Domain) are specifically designed to test whether models will refuse to provide dangerous advice or will correct misinformation in a user's query. The **RWE-LLM (Real-World Evaluation of LLMs in Healthcare)** framework proposes a comprehensive safety validation process inspired by red teaming, involving thousands of licensed clinicians to review model outputs for potential harm.

3) REGULATORY CONSIDERATIONS

As medical LLMs move closer to clinical practice, they will face regulatory scrutiny from bodies like the U.S. Food and Drug Administration (FDA). This introduces a new layer to evaluation, where models must not only be accurate and safe but must also meet formal standards for validation, documentation, and post-deployment monitoring. The development of standardized, regulatory-grade evaluation frameworks for medical AI is a critical and emerging field.

D. MULTIMODAL EVALUATION

The latest generation of LLMs are multimodal, capable of processing and reasoning about not just text but also images, audio, and video. This opens up new capabilities but also creates significant new evaluation challenges.

1) VISION-LANGUAGE BENCHMARKS

A wide array of benchmarks has been developed to evaluate vision-language models (VLMs). **MME (Multimodal Model Evaluation)** is a comprehensive benchmark that assesses both perception and cognition across 14 subtasks. **MM-Bench** uses carefully curated multiple-choice questions that require complex reasoning to answer from an image. Other benchmarks like **POPE** (Polling-based Object Probing Evaluation) are specifically designed to measure object hallucination, a common failure mode where models describe objects that are not present in an image.

2) AUDIO, SPEECH, AND VIDEO

Evaluation is also expanding to other modalities. Benchmarks such as **SUPERB** (Speech processing Universal Performance Benchmark) evaluate models across a wide range of speech processing tasks. For video benchmarks, **Video-MME** and **EgoSchema** test the ability to understand temporal events, object interactions, and causal relationships in video clips.

3) CROSS-MODAL CONSISTENCY

A key challenge in multimodal evaluation is assessing *cross-modal consistency*. A model should not only be able to describe an image but also answer questions about it coherently and reason about its contents without contradiction. Evaluating this deep, integrated understanding, rather than

just performance on isolated modal tasks, is a major focus of next-generation multimodal evaluation research [50].

VIII. ADVANCED EVALUATION PARADIGMS

In response to the limitations of traditional, static benchmarks, several advanced evaluation paradigms have emerged. These paradigms are characterized by their comprehensiveness, their focus on critical properties beyond accuracy, and

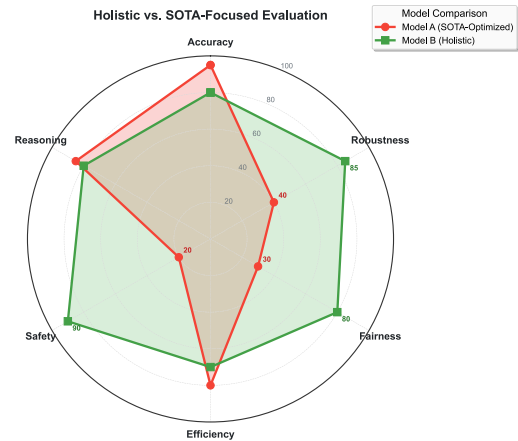


FIGURE 5. Holistic vs. SOTA-Chasing Performance. A visual comparison illustrating the “Evaluation-Capability Mismatch.” Model A achieves high accuracy but is brittle and unsafe. Model B, while having lower peak accuracy, offers a balanced profile suitable for real-world deployment.

their use of novel data sources and methodologies. They represent the frontier of evaluation science, aiming to provide a more holistic, reliable, and realistic assessment of LLM capabilities.

A. HOLISTIC EVALUATION FRAMEWORKS

Holistic frameworks are large-scale, multi-faceted evaluation suites that aim to assess LLMs across a broad spectrum of tasks and metrics simultaneously. Their goal is to move beyond single-score leaderboards to provide a more complete and transparent picture of a model's strengths. As shown in Figure 5, a holistic evaluation often reveals critical trade-offs between narrow accuracy and essential qualities, such as safety and robustness.

- HELM (Holistic Evaluation of Language Models):** Developed at the Stanford Center for Research on Foundation Models (CRFM), HELM is a pioneering effort in comprehensive evaluation [30]. It is built on a “taxonomy-down” approach, first defining the scenarios (e.g., question answering, summarization) and metrics (e.g., accuracy, robustness, fairness, bias, toxicity, efficiency) of interest, and then selecting representative datasets to cover this space. By evaluating dozens of models under standardized conditions, HELM enables direct, apples-to-apples comparisons and reveals important trade-offs, such as the fact that accuracy often does not correlate with robustness or fairness. The HELM framework is open-source and designed to

be a living benchmark, continuously updated with new models, scenarios, and metrics.

- **BIG-Bench (Beyond the Imitation Game Benchmark):** BIG-Bench is a massive, collaborative benchmark consisting of 204 tasks contributed by 450 authors [44]. Its primary design goal was to create tasks that are believed to be beyond the capabilities of current LLMs, thereby providing a long-lasting and challenging evaluation suite. The tasks are exceptionally diverse, covering everything from linguistics and commonsense reasoning to software development and physics [44]. A key focus of BIG-Bench is the study of **emergent abilities**, capabilities that appear to arise unpredictably at large model scales.
- **LMSYS Chatbot Arena:** While HELM and BIG-Bench are primarily static, the Chatbot Arena provides a dynamic, real-world evaluation platform [55]. It crowdsources human preferences by having users interact with two anonymous models simultaneously and vote for the better response. These pairwise comparisons are then used to calculate an Elo rating for each model, producing a live leaderboard that reflects real-world user preferences. This approach is particularly valuable for evaluating subjective qualities, such as helpfulness and conversational ability, which are difficult to capture with automated metrics.

While these holistic approaches offer a more comprehensive view of model performance, they are not without limitations. They are computationally expensive to run, and the choice of which scenarios and metrics to include is itself a subjective decision that can influence overall rankings. Table 9 summarizes the core dimensions commonly used in holistic evaluation frameworks, clarifying what each metric category is intended to capture beyond raw task accuracy.

TABLE 9. Holistic evaluation dimensions.

Dimension	What It Measures
Accuracy	Core task performance
Calibration	Confidence alignment with correctness
Robustness	Stability to input perturbations
Fairness	Equal performance across groups
Bias	Presence of harmful stereotypes
Toxicity	Generation of harmful content
Efficiency	Computational resource requirements

B. SAFETY AND TRUSTWORTHINESS EVALUATION

As LLMs become more powerful and widely deployed, ensuring their safety and trustworthiness is crucial. A dedicated subfield of evaluation has emerged to measure and mitigate potential harms, including the generation of misinformation, biased content, or toxic language.

- **Truthfulness and Honesty:** **TruthfulQA** is a benchmark specifically designed to measure whether a model

is truthful in answering questions where humans often hold false beliefs or misconceptions [32]. To perform well, a model must avoid repeating falsehoods learned from human text. The key finding from this work was that larger models were often *less* truthful, as they were better at mimicking common human errors [32].

- **Harmful Outputs:** Several benchmarks focus on evaluating a model's propensity to generate harmful content. **SafetyBench** is a comprehensive benchmark with over 11,000 multiple-choice questions covering seven categories of safety concerns, including illegal activities and offensiveness [54]. **HarmBench** provides a standardized framework for evaluating automated red-teaming techniques, assessing how effectively different methods can "jailbreak" models and elicit harmful behaviors [35].
- **Social Bias:** To measure and mitigate social biases related to gender, race, religion, and other demographics, specialized tools and datasets have been developed. Frameworks like **LangBiTe** provide a library of prompts designed to test for various biases like sexism, racism, and ageism. Other approaches, inspired by the Implicit Association Test from psychology, use prompt-based methods to uncover implicit biases that may not be apparent in standard tests.
- **Robustness:** A model's trustworthiness also depends on its robustness to small, often adversarial, changes in the input. **Robustness Gym** is an evaluation toolkit that unifies several paradigms for testing robustness, including evaluating on specific subpopulations, applying textual transformations (e.g., adding typos), and running adversarial attacks [18].

C. EFFICIENCY AND SUSTAINABILITY METRICS

For a long time, the primary focus of LLM development was on capability, with little regard for computational cost. As models have grown to trillions of parameters, efficiency and sustainability have become critical evaluation criteria, especially for practical deployment.

- **Inference Latency and Throughput:** For real-time applications, the speed of a model is crucial. Key metrics include **time-to-first-token** (how quickly the model begins generating a response) and **tokens per second** (the overall generation speed). These metrics are highly dependent on the hardware and software stack, making standardized reporting essential.
- **Memory Footprint:** The amount of memory required to run a model is a significant constraint. This includes the **model size** (the storage needed for the parameters) and the dynamic memory usage during inference, particularly the size of the **KV cache**, which stores attention keys and values and can grow very large for long contexts.
- **Energy Consumption and Carbon Footprint:** The environmental impact of training and running large

models is a growing concern. Metrics are being developed to quantify the **energy consumption** (measured in kilowatt-hours) and the associated **carbon footprint** (measured in tons of CO₂ equivalent) for both training and inference. This enables a more sustainable approach to AI development, favoring smaller and more efficient models whenever possible.

- **The Pareto Frontier:** Ultimately, there is a trade-off between a model's performance (e.g., accuracy) and its efficiency (e.g., latency, cost). A holistic evaluation should not just identify the "best" model on a single axis. However, it should map out the **Pareto frontier**, the set of models that offer an optimal balance, where no other model is both better and more efficient.

D. EMERGENT CAPABILITY DETECTION

One of the most fascinating and debated topics in LLM research is the phenomenon of "emergent abilities", capabilities that appear to arise suddenly and unpredictably in large models but are not present in smaller ones [7].

- **Defining and Detecting Emergence:** An ability is considered emergent if a model's performance on a task remains at or near random chance for smaller models and then suddenly spikes above random after a certain scale threshold is crossed. Detecting these abilities requires **scale-dependent evaluation**, where a family of models of different sizes is tested on the same benchmark to observe performance trends.
- **The Mirage Hypothesis:** A recent line of critical work has challenged the notion of emergence, proposing the "mirage hypothesis" [42]. This hypothesis argues that emergent abilities are not a fundamental property of the models themselves but are an artifact of the metrics used for evaluation. Specifically, nonlinear or discontinuous metrics (like exact-match accuracy) can transform a smooth, continuous improvement in a model's underlying capabilities into a sharp, seemingly emergent jump in performance. When a more continuous metric (like token edit distance or log-likelihood) is used, the "emergence" often disappears, revealing a predictable, gradual improvement with scale [42].

The debate over emergent abilities is central to the field of evaluation science. If emergence is real, it poses significant challenges for AI safety, as dangerous capabilities could appear without warning. If it is a mirage, it highlights the profound impact our choice of measurement can have on our scientific conclusions and underscores the need for more careful, principled metric selection.

IX. DISCUSSION AND IMPLICATIONS

A. THEORETICAL IMPLICATIONS

1) PARADIGM SHIFT IN EVALUATION SCIENCE

This survey's primary theoretical contribution is framing evaluation as a metrological science rather than a competitive benchmark race. This shift has several implications:

From Static to Dynamic Measurement: The Benchmark Lifecycle Framework (Section VI-A) challenges the prevailing assumption that benchmarks are static instruments. Viewing benchmarks as having distinct lifecycle phases, design, validation, deployment, monitoring, and retirement, establishes a theoretical foundation for understanding benchmark obsolescence as a natural and manageable occurrence rather than a crisis.

Construct Validity as Central Concern: The analysis reveals that the mismatch of evaluation-capability (Table 7) is fundamentally due to poor construct validity, benchmarks measuring proxies rather than target capabilities. This finding suggests that the field needs to adopt psychometric principles more rigorously, treating benchmark design as an exercise in psychological measurement rather than dataset curation.

2) GOODHART'S LAW IN AI EVALUATION

The analysis provides empirical support for Goodhart's Law operating in LLM evaluation. The documented divergence between benchmark scores and real-world performance across domains demonstrates that when metrics become targets, they cease to be good measures. This has profound implications for how we should structure research incentives and publication standards.

B. PRACTICAL IMPLICATIONS

1) FOR RESEARCHERS

The taxonomy (Section V) provides actionable guidance for evaluation selection. Rather than defaulting to popular benchmarks, researchers can now make principled choices based on explicit trade-offs between scalability, authenticity, and reproducibility (Table 4).

2) FOR PRACTITIONERS

The documented evaluation-capability gaps have immediate implications for deployment decisions. Organizations should not rely on benchmark scores as deployment criteria; instead, they should conduct domain-specific evaluations using real-world tasks. The medical domain analysis particularly highlights the risks of assuming benchmark performance translates to clinical competence.

3) FOR BENCHMARK CREATORS

The Lifecycle Framework provides a roadmap for responsible benchmark development. Key recommendations include:

- **Proactive Contamination Prevention:** Implement canary strings and versioning from inception
- **Retirement Planning:** Define clear obsolescence criteria at benchmark launch
- **Construct Validation:** Establish empirical correlation with real-world tasks before release

C. DIFFERENTIATION FROM EXISTING WORK

This survey differs from previous evaluation surveys in four key aspects:

1. Metrological Perspective: While surveys like Laskar et al. [28] catalog evaluation challenges, we uniquely frame evaluation through the lens of measurement science, introducing concepts like instrument calibration and lifecycle management.

2. Lifecycle Framework: No previous survey has proposed a systematic framework for benchmark lifecycle management. This novel contribution provides practical tools for the community.

3. Empirical Gap Analysis: The systematic documentation of benchmark-reality gaps (Table 7) with verified evidence goes beyond anecdotal observations to provide quantifiable (where possible) or well-documented qualitative evidence of the evaluation crisis.

4. Holistic Integration: Rather than focusing on specific aspects (e.g., Huang et al. 2024 on methods [21], Chen et al. 2025 on contamination [12]), we provide an integrated view connecting historical evolution, current pathologies, and future directions.

D. BROADER IMPACT ON AI DEVELOPMENT

The evaluation crisis documented in this survey has implications beyond academic research:

AI Safety: Misleading benchmark scores create false confidence in model capabilities, potentially leading to premature deployment in safety-critical applications. The analysis of medical benchmarks particularly highlights this risk.

Resource Allocation: The leaderboard obsession diverts resources toward marginal improvements on saturated benchmarks rather than addressing fundamental limitations. A metrological approach could redirect effort toward meaningful capability development.

Public Trust: The gap between advertised benchmark performance and real-world capability erodes public trust in AI. Transparent, lifecycle-managed evaluation could rebuild confidence through honest capability assessment.

X. THE FUTURE OF LLM EVALUATION

The field of LLM evaluation is at a critical inflection point. The limitations of the current paradigm are clear, and the need for a more scientific, robust, and reliable approach is urgent. This final section looks to the future, outlining the next generation of evaluation platforms, the growing importance of standardization and governance, key open research questions, and concrete recommendations for the community.

A. NEXT-GENERATION EVALUATION PLATFORMS

The static, one-off nature of traditional benchmarking is giving way to more dynamic, continuous, and automated evaluation platforms designed to keep pace with the rapid evolution of LLMs.

- **Continuous Evaluation (Eval-as-a-Service):** The future of evaluation in industry settings will likely

be continuous, integrated directly into the MLOps pipeline. Platforms like Humanloop, Galileo AI, and LangSmith offer “evaluation-as-a-service,” allowing developers to automatically run a suite of tests with every model update or prompt change. This brings the rigor of continuous integration/continuous deployment (CI/CD) from software engineering to LLM development, enabling real-time monitoring for performance regressions, safety violations, or data drift.

- **Federated Evaluation:** Evaluating models on sensitive, real-world user data without compromising privacy is a major challenge. **Federated Evaluation** offers a promising solution [49]. In this paradigm, evaluation is performed locally on a user’s device or a secure client-server. Instead of sending raw data to a central location, only the aggregated, anonymized evaluation results are shared. This approach, an extension of federated learning, could enable large-scale, privacy-preserving evaluation on realistic data distributions that are not publicly available [49].
- **Automated Benchmark Generation:** The manual creation of high-quality benchmarks is a significant bottleneck. A burgeoning research area is the use of AI itself to automate benchmark generation [17]. LLMs can be prompted to generate diverse and challenging test cases, including questions, code snippets, or reasoning problems. These generated instances can then be filtered and validated by other LLMs or human experts to ensure quality. This approach could dramatically accelerate the creation of new, customized, and contamination-resistant evaluation datasets.
- **Interactive Simulation Environments:** For evaluating agentic capabilities, the future lies in rich, interactive simulation environments. Beyond current web-based sandboxes like WebArena, we can expect the development of virtual worlds and complex, game-like environments where agents must perform long-horizon, multi-step tasks that require planning, memory, and adaptation. These platforms will allow for the evaluation of more general problem-solving abilities in controlled yet complex settings.

B. STANDARDIZATION AND GOVERNANCE

As LLMs become more integrated into society, evaluation will move beyond the research lab and become a matter of public policy and governance. This will necessitate a greater degree of standardization and oversight.

- **Evaluation Standards Bodies:** The current ad-hoc nature of benchmark creation is unsustainable. There is a growing need for independent, multi-stakeholder bodies to set standards for evaluation. Organizations like the U.S. National Institute of Standards and Technology (NIST) are already playing a key role in developing frameworks, such as the AI Risk Management Framework, and conducting evaluations of AI systems [1].

International standards bodies, such as the IEEE, are also developing standards for LLM evaluation [24].

- **Certification Frameworks:** In the future, high-stakes applications of LLMs (e.g., in medicine or finance) may require formal certification from trusted, third-party evaluators. These “AI auditors” would conduct rigorous, independent tests of a model’s performance, safety, and fairness against standardized criteria before it can be deployed.
- **Regulatory Requirements:** Governments are increasingly likely to mandate certain types of evaluation for AI systems. Regulations like the EU AI Act may require developers of high-risk AI systems to demonstrate compliance with safety and fairness standards through rigorous testing and documentation. This will transform evaluation from a research activity into a legal and regulatory necessity.
- **International Cooperation:** Given the global nature of AI development, international cooperation on evaluation standards will be crucial. Shared benchmarks and evaluation protocols will enable more meaningful comparisons of models developed in different countries, fostering a global consensus on what constitutes safe and trustworthy AI.

C. OPEN RESEARCH QUESTIONS

Despite rapid progress, many fundamental questions in the science of evaluation remain unanswered. These challenges will likely dominate the research agenda for the next five years:

- **Can we create contamination-proof evaluations?** With training data becoming ever larger, is it possible to design benchmarks that are fundamentally resistant to contamination? This may require new methods of data generation or evaluation protocols that test for generalization in ways that cannot be easily memorized.
- **How do we evaluate open-ended creativity?** Current evaluation methods are primarily focused on tasks with a single correct answer or a narrow range of acceptable outputs. How can we reliably and scalably measure the quality of creative outputs like writing a poem, composing music, or generating a novel scientific hypothesis?
- **What constitutes “understanding” vs. pattern matching?** This is a central philosophical and technical question in AI. Can we design evaluations that can definitively distinguish between a model that is only a “stochastic parrot” regurgitating statistical patterns and one that possesses a genuine, abstract understanding of the concepts it manipulates? This may require new paradigms that test for causal reasoning, counterfactual thinking, and systematic generalization [51].
- **How do we measure progress toward Artificial General Intelligence (AGI)?** Current benchmarks measure a collection of narrow skills. AGI implies a more

general, flexible, and autonomous intelligence. What would a true benchmark for AGI look like? How do we measure the ability to learn new skills, adapt to novel situations, and set one’s own goals? Answering this question is one of the grand challenges for the field [47].

D. RECOMMENDATIONS FOR THE COMMUNITY

To build a more rigorous evaluation culture, different stakeholders in the AI ecosystem must adopt new practices.

- **For Researchers:** Move beyond simply reporting SOTA scores. Conduct thorough error analyses. Report a wide range of metrics, including those for efficiency and fairness. Use statistical tests and report confidence intervals. When proposing new benchmarks, follow the principles of the Benchmark Lifecycle Framework.
- **For Practitioners:** Do not blindly trust leaderboard scores. Select benchmarks and evaluation methods that are directly relevant to your specific use case. Conduct your own internal evaluations on real-world data. Prioritize safety, robustness, and reliability over marginal gains in accuracy.
- **For Policymakers:** Understand the limitations of current evaluation methods. Do not rely on single benchmark scores as a basis for policy. Support the development of independent, public evaluation infrastructure and standards bodies. Mandate transparency in evaluation from developers of high-risk AI systems.
- **For Benchmark Creators:** Embrace the Benchmark Lifecycle Framework. Design for longevity by including challenging, multi-step reasoning tasks. Establish strong human baselines. Provide clear, standardized, versioned evaluation protocols. Proactively monitor for contamination and have a clear plan for retiring the benchmark when it becomes obsolete.

XI. CONCLUSION

The journey of Large Language Model evaluation has been one of rapid, reactive evolution. We have progressed from simple lexical-overlap metrics designed for narrow tasks to a sprawling, complex ecosystem of massive benchmarks, interactive environments, and holistic frameworks. This evolution, while impressive, has been driven by a frantic race to keep up with the exponential growth of model capabilities. The result is the current crisis of confidence: a landscape littered with saturated benchmarks, plagued by data contamination, and governed by the perverse incentives of the leaderboard. The scores are going up, but our understanding of what they mean is not. This survey aims to bring order and scientific rigor to this chaotic landscape. By tracing the historical arc of evaluation, we have shown how the field has repeatedly been forced to reinvent its measurement tools in response to disruptive new models. By introducing a systematic taxonomy of methodologies, we have provided a framework for understanding the fundamental trade-offs between scalability, reproducibility, and authenticity that govern all evaluation paradigms. Most critically, by

proposing the Benchmark Lifecycle Framework and advocating for a metrological mindset, we have charted a path forward, one that treats evaluation not as a game to be won, but as a science to be practiced. The key takeaways are clear. First, evaluation must be multi-dimensional; single accuracy scores are dangerously incomplete. We must measure not only what models can do, but also how robustly, fairly, and efficiently they do it. Second, data contamination is an existential threat to meaningful evaluation, and proactive detection and management must become standard practice. Third, every benchmark has a limited lifespan and must be managed accordingly, from its principled design to its planned retirement. Ultimately, the community must move beyond the SOTA race and cultivate a culture that values in-depth analysis, transparent reporting, and a focus on real-world utility over leaderboard rankings. The task ahead is not just technical; it is a challenge to the scientific culture of the AI community. It requires a collective commitment to rigor, transparency, and a shared understanding that a single number on a leaderboard does not measure genuine progress, but rather the creation of AI systems that are demonstrably reliable, safe, and beneficial for humanity. Building this more mature science of evaluation is the most critical and urgent task facing the field today.

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